

Indoor Navigation Using Wi-Fi Fingerprinting Combined with Pedestrian Dead Reckoning

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Abstract—This paper presents a method by which to calibrate the Wi-Fi fingerprinting database with less effort using smartphone based Pedestrian Dead Reckoning (PDR) without any a-priori knowledge. Because the accumulated errors from PDR will decrease the quality of the database, we employ a quaternion based orientation extended Kalman filter (EKF) to deal with the pedestrian heading and to narrow down the PDR positioning error to 2.2 meters for a 270 meter path. Furthermore, we implement the Walkie-Markie method to enhance the accuracy of the PDR step positions used to replace the reference points (RPs) built into the Wi-Fi fingerprinting database. The method defines the Wi-Fi landmarks according to the trend of the Receive Signal Strength (RSS) collected by PDR and converges the pathway map using the Wi-Fi Marks (WMs). In a simulation scenario, the WMs and PDR pathway errors is nearly 0 meters.

Keywords: *Wi-Fi fingerprinting, Pedestrian Dead Reckoning (PDR), Quaternion, EKF, Crowdsourced, Wi-Fi Mark.*

I. INTRODUCTION

Radio frequency (RF) fingerprinting based on Wi-Fi received signal strength (RSS) is a popular indoor navigation solution [1]. However, Wi-Fi fingerprinting must calibrate the database, which means receiving an RSS set at a specific reference point (RP). Pedestrian dead reckoning (PDR) along with the use of mobile built-in inertial measurement units (IMUs) and a magnetometer [2] can be used to reduce calibration effort [1, 3, 4]. Thus, the PDR position is obtained by accumulating the displacement of users and is annotated with the Wi-Fi RSS. Then, the reference points (RPs) used in Wi-Fi fingerprinting can be replaced by the users' step positions obtained by the PDR. Therefore, the PDR algorithm is implemented in different modes namely texting to receive the users' daily position data. Furthermore, the Walkie-Markie method based on the trend of Wi-Fi RSS [4] is applied to eliminate the cumulated error [2] from PDR and to build radio frequency map based Wi-Fi with a corrected PDR position as the RP.

In order to efficiently build the Wi-Fi database, the RSS database is conducted through crowdsourcing. That is, the PDR positioning information with the Wi-Fi RSS is collected from users [3]. We train the Wi-Fi data with the users' PDR pathway map, which must include the cumulated error caused by sensors. In this paper, we employ the Walkie-Markie method [4] to eliminate the cumulated PDR error. The method incorporate the landmark concept and define the Wi-Fi marks based on the Wi-Fi RSS trend. Using the Wi-Fi marks, the PDR position is pulled back once a mark is encountered. Because annotating the PDR

step positions with the Wi-Fi RSS is one of the Walkie-Markie process, we can contribute this information to the PDR calibrated Wi-Fi fingerprinting database. We then apply the K weighted nearest neighbor algorithm to perform the Wi-Fi fingerprinting positioning [1, 5].

We conducted PDR by fusing all the information from the IMU and magnetometer to acquire the displacement vector of the users. Since the displacement is derived from the step detection, stride length model, and heading information, we broke the PDR procedure into the following steps. First, to deal with the step detection, we applied an algorithm that chooses the step from the magnitude and the temporal phase of the acceleration to provide a step choice that is more robust against the dynamics of smartphone [6]. The stride length model was used to calculate the step length [7]. The heading information was obtained with a quaternion based extended Kalman filter (EKF) [8], which employs the concept of the quaternion preventing the singularity state. Also, the heading from the quaternion based EKF is more stable and reliable as compare to other orientation filters [8, 9]. We dealt with the sensor errors (offset and noise) using the previously referenced algorithm and filters, but there were still some cumulated errors in the PDR positioning phase had to be eliminated. Thus, we employ the Walkie-Markie method, as introduced previously.

In this paper, an indoor navigation algorithm is developed that allows the PDR to work with the cumulated error. The Walkie-Markie method not only eliminates the errors but also achieves the crowdsourcing for Wi-Fi database. To evaluate the accuracy of the PDR, experiments are conducted in a building with a smartphone held in texting mode. The Walkie-Markie method is then applied to correct the PDR positioning results. Subsequently, the Walkie-Markie can provide the information built into the Wi-Fi fingerprinting database. Furthermore, a comparison of the Wi-Fi fingerprinting results for the PDR-calibrated Wi-Fi database and the conventional Wi-Fi database can be made. It is expected that the comparison will show that the proposed methods, which uses a simpler procedure, can obtain the positioning performance as good as the conventional method.

II. METHODOLOGY

In order to conduct indoor navigation, both inertial sensing PDR [2] and Wi-Fi based fingerprinting localization [1] are the fundamental theories upon which this paper is based, where an indoor pathway mapping method called Walkie-Markie [4] is used to help us achieve our purpose.

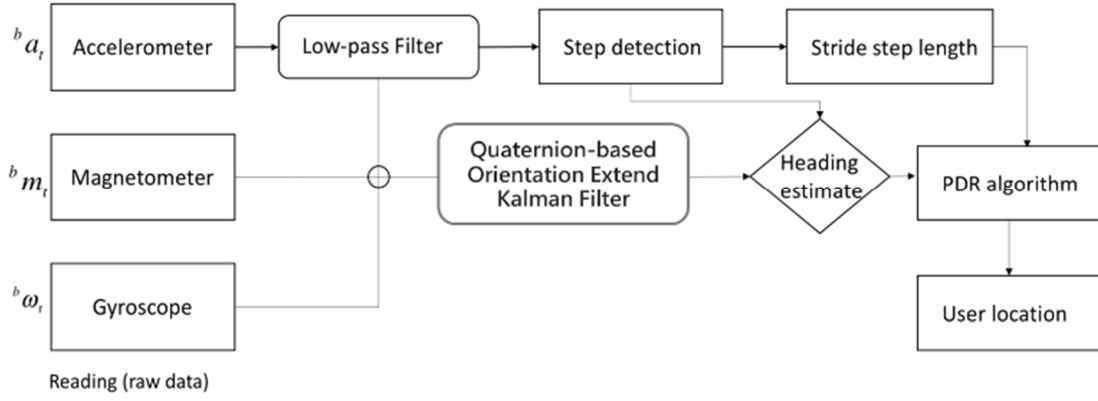


Figure 1 Flow diagram of PDR.

A. Pedestrian Dead Reckoning (PDR)

PDR can be divided into three parts: step detection, orientation, and stride length model [2]. Because PDR localization is based on the MEMS level inertial sensors built into a smartphone, the noise and bias from these sensors will cause errors. The discussion below explains the choice of algorithms and how we dealt such errors. The PDR flow diagram is shown as Figure 1.

1) Quaternion-based Orientation Extend Kalman Filter

Here, we first apply the sensor fusion concept by using several types of sensors to overcome the limitation of one type sensor. In this way, we employ the Kalman filter to make the best use of the sensor data. [8]

a) Kalman Filter Formulation

The EKF algorithm addresses the problem of estimating the state of a discrete-time process as described by the equations below:

$$x_{k+1} = A_k x_k + B_k u_k + w_k \quad (1)$$

$$z_{k+1} = H_k x_k + v_k, \quad (2)$$

Where k is the time samples, x is the state vector, u is the input vector, A , B , H are the state, input, and output matrix, and w , v are the state and measurement noise, respectively.

b) Quaternion-Based EKF

The orientation was mainly obtained from integrating the angular velocity, but it still suffered from the gyro bias error. We used the angular velocity to estimate the orientation state and treated Earth's gravitational and magnetic field vectors as the measurement to bound the bias. The quaternions were used to improve computation efficiency and avoid singularities.

The continuous-time nonlinear system equations are

$$\dot{x} = f(x, \omega) + w \quad (3)$$

$$y = h(x) + v, \quad (4)$$

where $x = [q_0 \ q_1 \ q_2 \ q_3 \ b_{\omega x} \ b_{\omega y} \ b_{\omega z}]^T$ is the system state vector, and $y = [a_x \ a_y \ a_z \ m_x \ m_y \ m_z]$ is the measurement

vector. The symbol q is the quaternions, b is the gyro bias, a is the acceleration, and m is magnetic field.

Here, we want to linearize this system, so we choose state model A and measurement model H by calculating the Jacobian of $f(x, \omega)$ and $h(x)$ as follow:

$$F = \frac{\partial f(x, \omega)}{\partial x} \rightarrow A = I + F * \Delta t \quad (5)$$

$$H = \frac{\partial h(x)}{\partial x} \quad (6)$$

And

$$f(x, \omega) = \begin{bmatrix} \frac{1}{2} \begin{bmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & q_0 & -q_1 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix} \begin{bmatrix} \omega_x - b_{\omega x} \\ \omega_y - b_{\omega y} \\ \omega_z - b_{\omega z} \end{bmatrix} \\ 0^{3 \times 1} \end{bmatrix} \quad (7)$$

$$h(x) = \begin{bmatrix} C_n^b(q) & 0 \\ 0 & C_n^b(q) \end{bmatrix} \begin{bmatrix} \vec{g} \\ \vec{m} \end{bmatrix}, \quad (8)$$

where $C_n^b(q)$ is the direct cosine matrix, which represent the coordinate transformation from the navigation frame to the body frame. $\vec{g}$ is the gravity vector, and $\vec{m}$ is the Earth's magnetic field. Both them are in the NED frame.

With these linearized equations, the orientation problem can be optimized by the extended Kalman filter (EKF).

2) Robust Step Detection Algorithm

The robust step detection algorithm [6] was based on the accelerometer data. The low pass filter was used to filter out the noise in the accelerometer and to make the acceleration trend of more obvious. This algorithm detects steps from the magnitude phase and the temporal phase. Assuming the neighbor peaks or valleys will not show in a very short period, it counts a step when a pair of peaks and valleys appears. For example, assume there are two decided peaks appearing continuously. The algorithm will treat one as fake if the between-temporal is lower

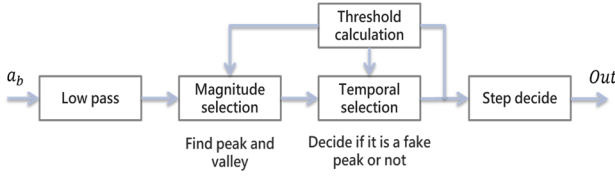


Figure 2 Flow diagram of robust step detection algorithm.

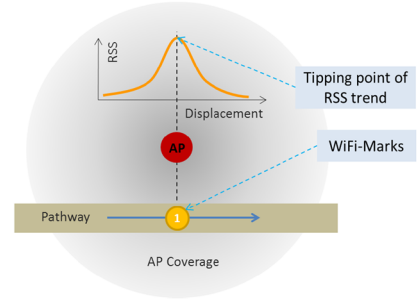


Figure 3 Illustration of Wi-Fi Marks

3) The Stride Length Model

Users may walk using different stride lengths every step that results in large or small acceleration. We must clarify this by making the stride length model be sensitive to changes in acceleration. We chose a model related to the standard deviation of the acceleration data [7].

$$SL = \alpha \times \sigma_a + \beta, \quad (9)$$

where SL is the stride length, σ_a is the standard deviation of the acceleration, and α and β are the parameters in this model.

$$W_j = \frac{1/d_j}{\sum_{j=1}^K 1/d_j}, \quad (X, Y) = \sum_{j=1}^K (W_j * (X_j, Y_j)), \quad (10)$$

where N is the total number of APs annotated with this RP, $rss(pos, i)$ is the RSS from the i_{th} AP, $rss(j, i)$ is RSS from the i_{th} AP annotated with the j_{th} RP, (X_j, Y_j) is the position of the j_{th} RP, K is the quantity of RPs that we want to use to weight.

B. Walkie-Markie Method

Landmarks is used to give directions. Regardless of how one detours, once a landmark is encountered, previous errors are reset. This idea can be used to limit the drift and cumulated error in the PDR and align the trajectory of different users. Using the concept, this method identify landmark based on Wi-Fi signal since nowadays smartphone can sensing the Wi-Fi signal in the background. This method called it as Wi-Fi Marks. [4]

1) Wi-Fi Marks

a) Concept

In Wi-Fi fingerprinting, we use the Wi-Fi RSS value directly, but this method pays attention to the trend of the RSS changes. The trend of the RSS is related to the distance between the user and the true APs. The basic idea is shown in Figure 3. If a user walks closer and through the point that has the shortest distance to an AP, the RSS trend will first be an increase then a decrease. The point that has the maximum RSS value is called RSS trend tipped point (RTTP), which should occur around the same location. These RTTPs are treated as the Wi-Fi Marks candidate.

b) Identification

As a landmark, each Wi-Fi Mark should be uniquely identifiable. Although these RTTPs can be simply distinguished by the BSSID (the MAC address of the APs), more information is needed to cluster these candidates of Wi-Fi marks since one AP may generate multiple RTTPs depending on how the RSS coverage area intersects with the pathway as shown in Figure 4. Thus, the Walkie-Markie method annotates these Wi-Fi Mark candidates with the following four-tuple:

$$WM \triangleq \{BSSID, pos, (D_1, D_2), \mathcal{N}\}, \quad (11)$$

where $BSSID$ is the ID of the master AP, pos is the PDR-estimated position of this detected RTTP, D_1 and D_2 are the steady walking PDR-estimated directions approaching and leaving the RTTP, respectively, $\mathcal{N}$ is the set of information for the neighboring APs, including their respective RSS differences from the master AP. The following shows how the method cluster different Wi-Fi Marks from the same master AP.

There is a scenario shown in Figure 5 where a user encounters multiple RTTPs that are coming from the same master AP when the user walks on the pathway. Using directions D_1 and D_2 , RTTP 1, {2, 3}, 4, 5 can be easily differentiate. Moreover, the U-turn situation (RTTP 6 and 7) can be detected, but they are treated as erroneous RTTP because it may add significant noise to the system.

If the RTTPs show in parallel corridors or exhibit similar turning styles, like {2, 3}, they may have similar D_1 and D_2 . In this situation, $\mathcal{N}$ is helpful to differentiate among these

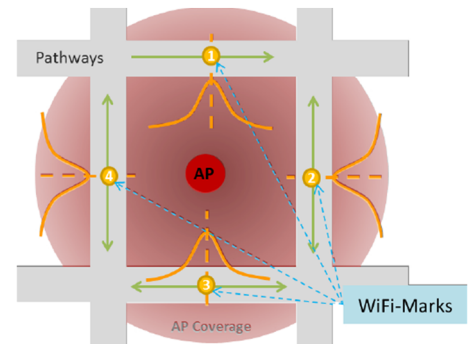


Figure 4 Possibly multiple Wi-Fi Marks for the same AP.

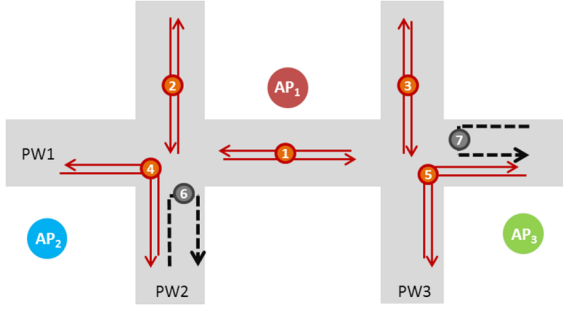


Figure 5 Multiple RTTP possibilities for AP1 under different walking patterns illustrated by arrows

RTTPs. If the difference between the $\mathcal{N}$ of two RTTPs is small, then these two RTTPs are exactly the same Wi-Fi Mark. Under these conditions, the Wi-Fi Marks clustering process can be implemented.

2) Arturia Positioning Algorithm

The major purpose of this method is to limit the drift and cumulated error in the PDR using the landmark concept. From the previous work, we have already defined the Wi-Fi Marks that are ready for use. Arturia is a mapping algorithm based on the spring relaxation concept (Figure 6). It treats each edge of the graph as a spring, where the entire graph forms a spring network.

a) Building the Graph

To build the spring network, the method defines the Wi-Fi Marks as the nodes and each edge that is the actual displacement vector from the user measured by PDR is added between two specific nodes. It is possible that there exists more than one edge between a pair of nodes when there are multiple user trajectories. Thus, this method utilizes for these PDR-estimated constraints to make the Wi-Fi Marks coordinate more correct.

b) Realizing the Graph

The goal of the Arturia algorithm is to minimize the potential energy E which is the function of the discrepancy between the calculated displacement vectors (treated as the actual length of the springs) and the measurement constraints (treated as the rest length of the springs). The solution of this algorithm involves adjusting the node position as if it is pushed or pulled by

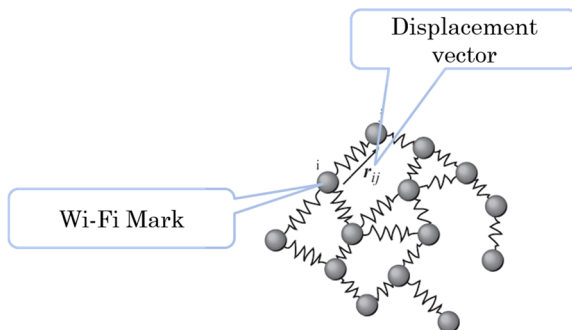


Figure 6 Schematic of the Arturia algorithm

the net force from all the other connected neighboring springs. Following are the Arturia steps.

First, we simply assume the coordinate of the nodes are at the origin and that these coordinates will be updated by the incoming data. Next, the iteration is applied to minimize the potential energy E . The equations to calculate E can be expressed as follows:

$$\tilde{\epsilon}_{i,j} = \sum_{k=1}^{N_{e,i,j}} (\tilde{r}_{i,j,k} - \tilde{d}_{ij}) \quad (12)$$

$$\tilde{F}_i = \frac{1}{N_{e,i}} \sum_j \tilde{\epsilon}_{i,j} \quad (13)$$

$$\hat{p}_i = \hat{p}_i + \tilde{F}_i \quad (14)$$

$$E_i = \sum_j |\tilde{\epsilon}_{i,j}|^2, E = \sum_i E_i, \quad (15)$$

where $\hat{p}_i$ is the current coordinate of the i_{th} node, $\tilde{d}_{ij} = \hat{p}_i - \hat{p}_j$ is the current displacement between the i_{th} node and the neighboring j_{th} node, $N_{e,i,j}$ is the number of real measurement constraint between the i_{th} and j_{th} nodes, $\tilde{r}_{i,j,k}$ is the k_{th} constraint, $\tilde{\epsilon}_{i,j}$ is the difference between the current displacement vectors and the constraints, $\tilde{F}_i$ is the adjustment of the i_{th} node and $N_{e,i}$ is the total number of constraints of the i_{th} node.

The coordinates of the nodes are updated in Equation (14), thus making $\tilde{\epsilon}$ smaller in order to minimize the potential energy E . When E is smaller than a selected threshold, Arturia stops the iteration until it obtains information for a new incoming node (Wi-Fi Mark). More information from users will make the pathway map more develop.

After we obtain reliable pathway map as built by Walkie-Markie, the estimated-PDR position will be corrected and thus can be used as the RPs to build the radio frequency map used as the Wi-Fi database mentioned previously

C. Radio Frequency Fingerprinting based on Wi-Fi

The Wi-Fi fingerprinting method can be divided into 2 steps: calibration step and matching and positioning step. The flow diagram is shown in Figure 7. In the calibration step, we

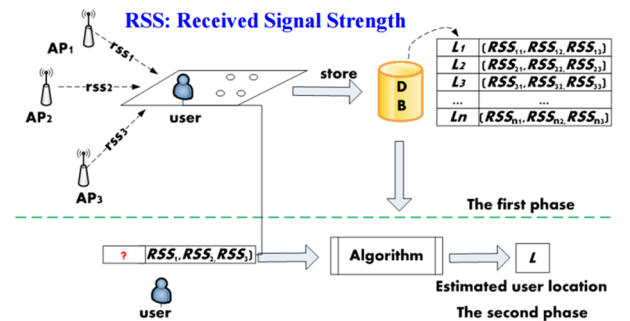


Figure 7 Flow diagram of Wi-Fi fingerprinting

must receive the RSS from all access points (APs) at specific reference points (RPs) to build the Wi-Fi database. In the matching and positioning step, when a user obtains a Wi-Fi RSS sequence, the algorithm will match it with the database and estimate the user's position. And we chose the K weighted near neighbor (KWNN) method as the matching algorithm [1] :

$$d_j = \frac{1}{N} \left[\sum_{i=1}^N |rss(pos, i) - RSS(j, i)|^2 \right]^{\frac{1}{2}} \quad (16)$$

$$W_j = \frac{1/d_j}{\sum_{j=1}^K 1/d_j}, \quad (X, Y) = \sum_{j=1}^K (W_j * (X_j, Y_j)), \quad (17)$$

where N is the total number of APs annotated with this RP, $rss(pos, i)$ is the RSS from the i_{th} AP, $RSS(j, i)$ is RSS from the i_{th} AP annotated with the j_{th} RP, (X_j, Y_j) is the position of the j_{th} RP, K is the quantity of RPs that we want to use to weight.

III. EXPERIMENTAL RESULTS

In order to ensure the quality of the Wi-Fi database when using PDR to calibrate it, we did a series of experiments to test the methods mentioned earlier before and to determine: (a) how the result of PDR positioning effected by the cumulated error heading and the steps with stride length counted by noisy acceleration, (b) whether the Walkie-Markie method can be used to deal with the PDR error, (c) if the reliability of the pathway map built by Walkie-Markie is adequate to be the RPs and calibrate the database used by Wi-Fi fingerprinting. In addition, we conducted some tests on the KWNN RSS matching algorithm to show that we were ready for the PDR-calibrated database to perform the PDR-calibrated Wi-Fi fingerprinting.

A. PDR Evaluation

We divide it into two parts to evaluate the performance of the PDR algorithm: the orientation experiment the PDR positioning result. Both them were tested on actual walking data.

1) Pedestrian Heading

The first problem we encountered in PDR is that the heading obtained from the sensor data and may suffer from a drift heading error caused by sensor errors. Quaternion-based orientation EKF was aimed toward solving this problem, as mentioned previously. Here, we used actual walking data to test this EKF algorithm. This data was received by a Samsung Galaxy S8 with the user holding the smartphone in texting mode. The user walked in a square for 2 rounds on the 3rd floor of the Department of Aeronautics and Astronautics (DAA). The results are shown in Figure 8 and TABLE I. Notice that we compare the EKF results to the other results from an algorithm called the quaternion complementary orientation filter (QCF), which can obtain orientation information from the data from 3-axis IMUs with a 3-axis magnetometer.

TABLE I shows that the standard deviation and the mean of the heading error calculated using the EKF were both smaller than the others obtained by QCF. This indicates that the EKF has more accurate and convergent heading performance. The same conclusion is shown in Figure 8, where the blue data provided by the EKF seems to be approaching to the true heading that painted in black. Thus, the quaternion-based orientation EKF was applied to do the subsequent experiments.

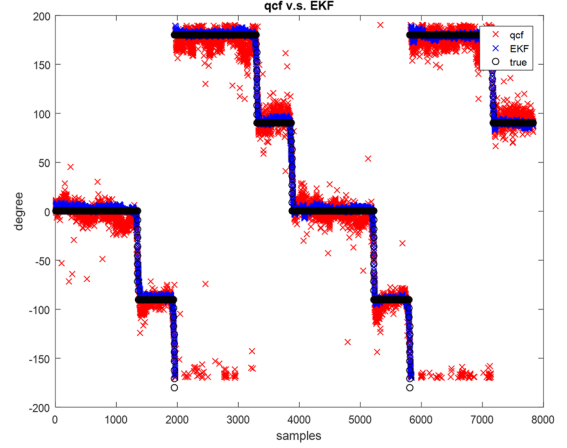


Figure 8 The heading result from EKF vs. QCF.

TABLE I HEADING ERROR COMPARISON

Filter	Mean	Std.
Quaternion Complementary Filter	-9.17 deg.	7.99 deg.
Quaternion Extend Kalman Filter	-0.3212 deg.	1.93 deg.

2) PDR Positioning

We combined the pedestrian heading with the step detection algorithm and the stride length model to implement the PDR positioning. We tested this PDR method in 2 cases: (a) walking in a square shape for 3 rounds for which each round was 90 meters long in the 3rd floor of the DAA, and the PDR positioning result was recorded for every 10 meters and (b) walking in a hourglass shape for 5 rounds for which each round is 30.1 meters in the 1st floor of DAA (Figure 9), and the PDR positioning result was recorded for every turning point. The testing phone was a Samsung Galaxy S8. The user was holding the smartphone in texting mode and walking in the shapes mentioned above. We plotted the result as a cumulative-distribution-function (CDF) where the x-axis was the positioning error between the recorded points and the true points, so we could see

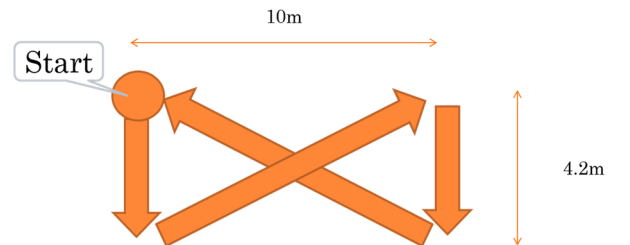


Figure 9 Schematic of the hourglass shape walking

the cumulated error in PDR directly. The results are shown in Figure 10 and Figure 11 and in TABLE II and TABLE III. In a comparison of both these experiments, all the turning angles in condition (b) were found to be greater than those in condition (a), but the positioning performance appeared to be the same. From these results, it could be seen that the proposed PDR algorithm had approximately 2 meters cumulated error for a 150-meter walk.

a) Walks in a square pattern for 3 rounds (270 m)

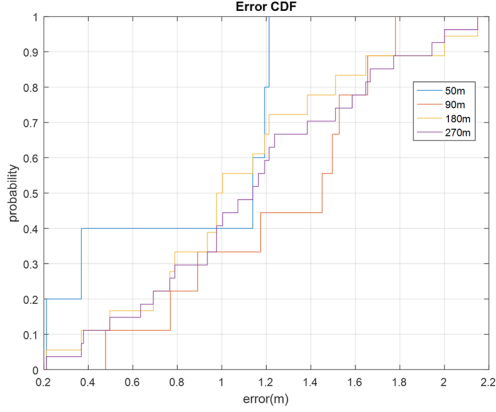


Figure 10 The CDF of the cumulated error in PDR (square shape)

TABLE II POSITIONING ERROR W.R.T. THE WALKING DISTANCE

Distance	50 m	90 m	180 m	270 m
Error mean (m)	0.8246	1.2469	1.0804	1.1445
Error STD (m)	0.4922	0.4454	0.5207	0.5206

b) Walks in a hourglass pattern for 5 rounds (150.5 m)

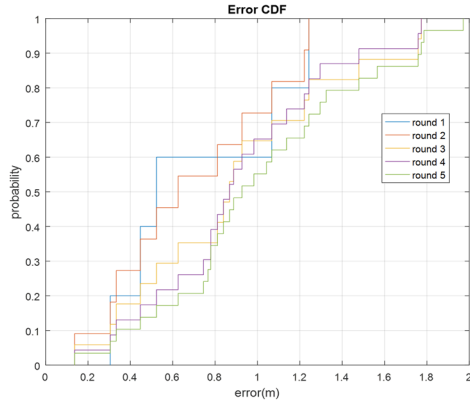


Figure 11 The CDF of the cumulated error in PDR (hourglass shape)

TABLE III POSITIONING ERROR W.R.T. THE WALKING ROUND(S)

Rounds	1	1~2	1~3	1~4	1~5
Error mean (m)	0.7170	0.6946	0.8968	0.9116	1.0146
Error STD (m)	0.4123	0.3838	0.4859	0.4289	0.4729

Then, we employed the Walkie-Markie not only to correct the cumulated error in the PDR but also to calibrate the Wi-Fi fingerprinting database using the step position from the PDR positioning result.

B. Walkie-Markie Method Simulation

In this section, we simulate the Walkie-Markie method to ensure that this method can help deal with the cumulated error in PDR positioning. In our simulation, we set a total of 10 Wi-Fi Aps on the outside of the pathway and assumed that the Wi-Fi RSS trend would be stable enough for the user to receive a set of pseudo-RSSs transferred from all Aps when steps occurred. The basic setting is shown in Figure 12.

The pseudo-RSS is assumed to be related to the distance between APs and the PDR step positions as in the following formula:

$$\text{pseudo-RSS} = \frac{1}{(\text{Distance}_{AP-PDR_Pos})^{0.25}} \cdot \quad (18)$$

In addition, we designed two different scenarios to test the method: (a) a pathway mapped by PDR without cumulated error and (b) a pathway mapped by PDR with cumulated error. Both the pathways are in the shape of square and are walked by a user with the same step length of 0.66 of a meter. The user walks 3 rounds for which each round has 80 steps. In the case with cumulated error, the PDR positioning result starts to drift for 0.03 of a meter in both the x and y coordinates for each step since the second round. The schematics of both scenarios are shown in Figure 13 and Figure 14, respectively.

These simulation results show in Figure 15 and Figure 16 that present the positioning error for the first case, where the PDR without cumulated error is 0 meters both before and after using the Walkie-Markie to correct it. This shows that the Walkie-Markie will not cause other errors when the PDR information and the Wi-Fi signal both lack noise. For the results of the second case where the PDR has cumulated error, we find that when the Wi-Fi signal does not have noise, and that the Walkie-Markie method can correct the cumulated error in the PDR by applying the Arturia algorithm to obtain the Wi-Fi Mark position and calibrate the PDR pathway.

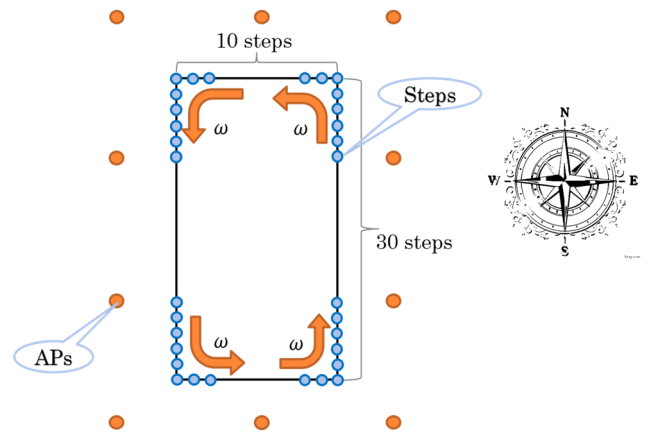


Figure 12 The basic setting of the Walkie-Markie simulation

The only uncertainty in these simulations is the stability of the actual Wi-Fi RSS trend. Since it is difficult to simulate the noise in a Wi-Fi RSS, we did the following experiment to test the stability of the Wi-Fi RSS trend to ensure that the Walkie-Markie method could be implemented.

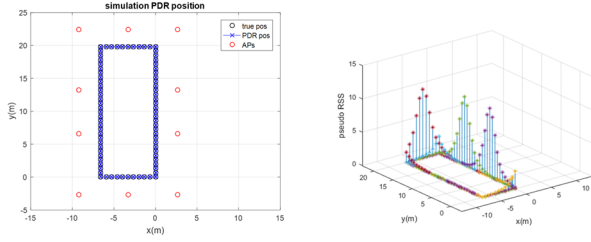


Figure 13 The simulated position of without- error-PDR and Wi-Fi Aps and the pseudo-RSS trend from number 1 to 5 Wi-Fi APs

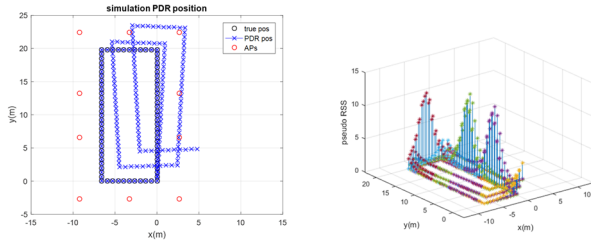


Figure 14 The simulated position of with- error-PDR and Wi-Fi AP and the pseudo-RSS trend from number 1 to 5 Wi-Fi AP

a) The simulation results for the PDR without error

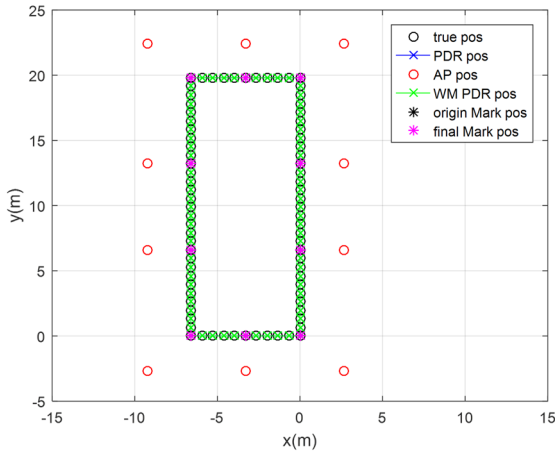


Figure 15 Walkie-Markie simulation result for without-error-PDR

TABLE IV COMPARISON OF THE POSITIONING ERROR FOR THE ORIGINAL PDR AND THE WALKIE-MARKIE CALIBRATED PDR

Positioning error	Mean	Std.
Original	0 m	0 m
Walkie-Markie	0 m	0 m

b) The simulation results for the PDR with error

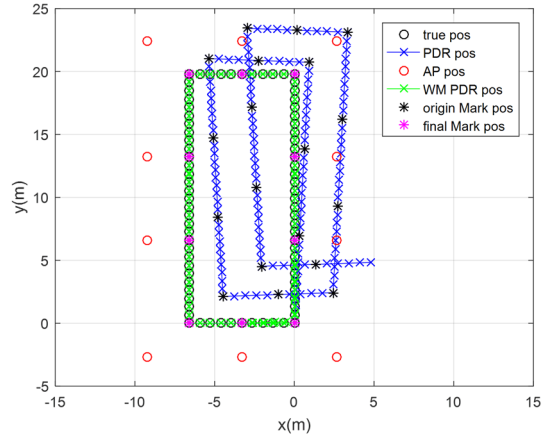


Figure 16 Walkie-Markie simulation result for with-error-PDR

TABLE V COMPARISON OF THE POSITIONING ERROR FOR THE ORIGINAL PDR AND THE WALKIE-MARKIE CALIBRATED PDR

Positioning error	Mean	Std.
Original	2.3053 m	2.2886 m
Walkie-Markie	0.0044 m	0.0167 m

C. Wi-Fi RSS Trend Stability Experiment

We designed an experiment to test whether the trend for the Wi-Fi RSS is stable enough for the Walkie-Markie. We let a user walk on a 32 meters long corridor in the DAA back and forth and erected a Wi-Fi AP in the middle (16 meters away from the starting point) of the corridor. A smartphone held by the user in texting mode gave us the user's PDR positioning result and the Wi-Fi RSS information from the erected AP synchronously. The user walked through the AP 5 times. The Wi-Fi RSS trend for this AP is shown in Figure 17.

The results show that the peak of the Wi-Fi RSS trend to be regular at around 15 to 17 meters. The position of the peak has about a 1-meter error are (a) the noise in the Wi-Fi RSS was not filtered and (b) the PDR positioning result was not correct. However, the results proved that the stability of the Wi-Fi RSS trend was adequate for the Walkie-Markie.

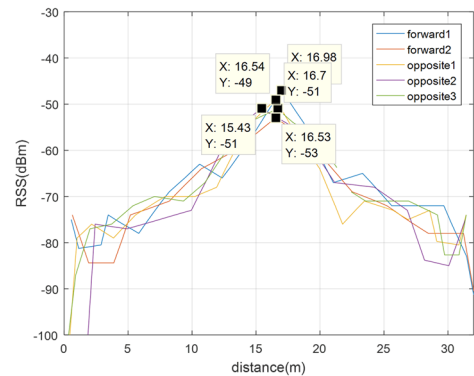


Figure 17 The Wi-Fi RSS trend from erected AP

D. KWNN Matching Algorithm for Wi-Fi Fingerprinting

We chose KWNN algorithm as our matching algorithm in Wi-Fi fingerprinting and conducted an experiment to test. We had already calibrated the Wi-Fi fingerprinting database for 81 RPs in a square shape pathway map, and we took one unit of data as the user receiving Wi-Fi RSS for every 4 units of data. Thus we obtained the database for 61 RPs and 20 user Wi-Fi RSSs in the experiment. The KWNN positioning results are plotted in CDF and are shown in Figure 18 and TABLE VI. We chose different K to achieve the best performance, but the results are similar to each other when K is in a range from 3 to 6.

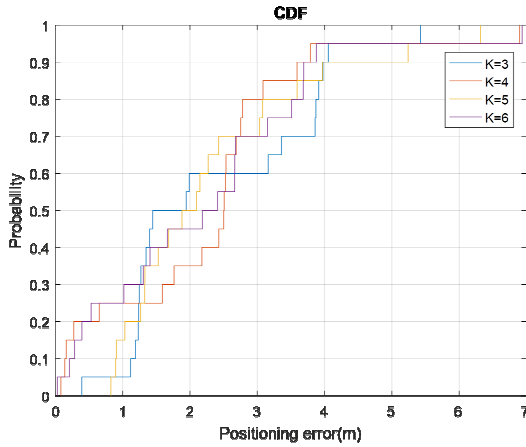


Figure 18 The KWNN positioning error CDF for different K

TABLE VI THE POSITIONING ERROR FOR KWNN ALGORITHM WITH DIFFERENT K

K	3	4	5	6
Error mean(m)	2.3720	2.2467	2.3426	2.2149
Error STD(m)	1.4194	1.5918	1.4906	1.6932

IV. CONCLUSION

The object of this work is to use the PDR positioning result to calibrate the Wi-Fi database, we focused on minimizing the sensor error and on finding a way to limit the cumulated error in PDR using Wi-Fi. The experimental results ensure that the cumulated error in PDR is no more than 2.2 meters for a 270-meter pathway when the phone is held in texting mode. Moreover, the Walkie-Markie based on the Wi-Fi RSS trend is efficient for eliminating the cumulated error in PDR in our simulated situation. On the other hand, the results shown for the Wi-Fi stability experiment also improve the feasibility of Walkie-

Markie for real Wi-Fi information. Also, the KWNN matching algorithm is sufficient for the PDR calibrated Wi-Fi database to perform the PDR-calibrated Wi-Fi fingerprinting.

After implementing the Walkie-Markie method for a real walk, every corrected step positions counted using PDR which built into the pathway map had to be annotated with a set of Wi-Fi RSSs. This information is the fundamental principle of the PDR calibrated Wi-Fi fingerprinting database. Building and testing the quality of the database will be the next step in our road map.

V. REFERENCES

- [1] S.-S. JAN, L.-T. HSU and W.-M. TSAI, "Development of an indoor location based service test bed and geographic information system with a wireless sensor network.," in *Sensors*, 2010.
- [2] W. KANG and Y. HAN, "SmartPDR: Smartphone-Based Pedestrian Dead Reckoning for Indoor Localization," in *IEEE SENSORS JOURNAL*, 2014.
- [3] A. e. a. RAI, "Zee: Zero-effort crowdsourcing for indoor localization.," in *Proceedings of the 18th annual international conference on Mobile computing and networking.*, 2012.
- [4] G. e. a. SHEN, "Walkie-markie: indoor pathway mapping made easy.," *Proceedings of the 10th USENIX conference on Networked Systems Design and Implementation.*, pp. 85-98, 2013.
- [5] M. YOUSSEF and A. AGRAWALA, "The Horus WLAN location determination system.," *Proceedings of the 3rd international conference on Mobile systems, applications, and services*, pp. 205-218, 2005.
- [6] H.-h. LEE, S. CHOI and M.-j. LEE, "Step detection robust against the dynamics of smartphones," in *Sensors*, 2015.
- [7] A. ERIC, "Motion Classification and Step Length Estimation for GPS/INS Pedestrian Navigation.," Stockholm, Sweden, 2012.
- [8] C. M. BRIGANTE, "Towards miniaturization of a MEMS-based wearable motion capture system.," in *IEEE Transactions on Industrial Electronics*, 2011.
- [9] R. G. VALENTI, I. DRYANOVSKI and J. XIAO, "Keeping a good attitude: A quaternion-based orientation filter for IMUs and MARGs.," in *Sensors*, 2015.
- [10] Y. KIM, Y. CHON and H. CHA, "Smartphone-based collaborative and autonomous radio fingerprinting.," in *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, 2012.
- [11] Z.-A. e. a. DENG, "Extended kalman filter for real time indoor localization by fusing WiFi and smartphone inertial sensors.," in *Micromachines*, 2015.